

Select: σ

↳ Used to select a subset of tuples based on a selection condition.

$$\sigma_{\langle \text{selection-condition} \rangle} (R)$$

↳ Is commutative

$$\begin{aligned} \sigma_{\text{and-1}} (\sigma_{\text{and-2}} (R)) &= \sigma_{\text{and-2}} (\sigma_{\text{and-1}} (R)) \\ &= \sigma_{\text{and-1 AND and-2}} (R) \end{aligned}$$

Project: π

↳ Keeps certain columns from a relation & discards the other columns

$$\pi_{\langle \text{attribute-list} \rangle} (\text{Employee})$$

↳ Removes any duplicate tuples

$$\pi_{\text{Fname, Lname, salary}} (\sigma_{\text{DNO}=5} (\text{Employee}))$$

$$\text{Dep5_Emps} \leftarrow \sigma_{\text{DNO}=5} (\text{Employee})$$

$$\text{Result} \leftarrow \pi_{\text{Fname, Lname, salary}} (\text{Dep5_Emps})$$

Select Fname, Lname,
Salary From Employees
Where DNO=5;

Rename: ρ

$\rho_S(B_1, B_2, \dots, B_n) (R) \rightarrow$ Change Relation name to S & attributes to $B_1, \dots, B_n$

$\rho_S (R) \rightarrow$ Change relation name only

$\rho_{(B_1, B_2, \dots, B_n)} (R) \rightarrow$ Change attribute name only

Union: $\cup$

$R \cup S \rightarrow$ Includes tuples that are either in R or in S

↳ R & S must have same number of attributes/names

Example,

$$\text{Dep5_Emps} \leftarrow \sigma_{\text{dno}=5}(\text{Employee})$$

$$\text{Result1} \leftarrow \pi_{\text{ssn}}(\text{Dep5_Emps})$$

$$\text{Result2}(\text{SSN}) \leftarrow \pi_{\text{superssn}}(\text{Dep5_Emps}) \leftarrow \text{Name of new col is ssn}$$

$$\text{Result} \leftarrow \text{Result1} \cup \text{Result2}$$

Intersection: $\cap$

Difference: $-$

Cartesian Product: $\times$

$$Q(A_1, A_2, \dots, A_n, B_1, B_2, \dots, B_m) = R(A_1, A_2, \dots, A_n) \times S(B_1, B_2, \dots, B_m)$$

$n_R * n_S$ tuples

n_R tuples

n_S tuples

Join:

$$R \bowtie_{\langle \text{join-condition} \rangle} S$$

$$\text{Dept_mgrs} \leftarrow \text{Department} \bowtie_{\text{mgrssn}=\text{ssn}} \text{Employee}$$

Natural Join: $*$

$$Q \leftarrow R(A, B, C, D) * S(C, D, E) \rightarrow Q(A, B, C, D, E)$$

$\hookrightarrow$ Implicit join condition created $R.C = S.C$

Example,

$$\text{Dept_locs} \leftarrow \text{Department} * \text{Dept_locations}$$

Left Join: $\bowtie_{\text{left}}$

Right Join: $\bowtie_{\text{right}}$

Full outer Join: $\bowtie_{\text{full}}$

Theta Join: $R \bowtie_{x > y} S$

Equi Join: $R \bowtie_{x=y} S$

Aggregation function:

$f_{\text{aggregate-func attribute}}(R)$

Example,

$f_{\text{max salary}}(\text{Employee})$

Book Qs:

Q16.

(a) $\text{Emp-dept5} \leftarrow \sigma_{\text{DNO}=5}(\text{Employees})$

$\text{Works-on-10} \leftarrow \sigma_{\text{Hours} > 10}(\text{Works-on})$

$\text{Emp-dept5-proj} \leftarrow \sigma_{\text{Pname} = \text{'ProductX'}}(\text{Emp-dept5} \bowtie_{\text{DNO} = \text{DNO}} \text{Project})$

$\text{Result} \leftarrow \pi_{\text{Fname, Lname}}(\text{Emp-dept5-proj} \bowtie_{\text{Pno} = \text{Pnumber}} \text{Works-on-10})$

(b) $\text{Emp-Dep} \leftarrow \text{Employees} \bowtie_{\text{SSN} = \text{ESSN}} \text{Dependent}$

$\text{Emp-Dep-same} \leftarrow \sigma_{\text{Fname} = \text{Dependent-name}}(\text{Emp-Dep})$

$\text{Result} \leftarrow \pi_{\text{Fname, Lname}}(\text{Emp-Dep})$

(c)

$\text{Sup-FW(SSSN)} \leftarrow \pi_{\text{SSN}}(\sigma_{\text{Fname} = \text{'Franklin'} \text{ AND } \text{Lname} = \text{'Wong'}}(\text{Employees}))$

$\text{Result} \leftarrow \pi_{\text{Fname, Lname}}(\text{Employees} \bowtie_{\text{SSN} = \text{SSSN}} \text{Sup-FW})$

$$(e) \text{Working-emp} \leftarrow \rho_{\text{ssn}}(\pi_{\text{ssn}}(\text{Works-ON}))$$

$$\text{No-proj-emp} \leftarrow \pi_{\text{ssn}}(\text{Employee}) - \text{Working-emp}$$

$$\text{Result} \leftarrow \pi_{\text{Fname, Lname}}(\text{Employee} \bowtie_{\text{no-proj-emp.ssn} = \text{Employee.ssn}} \text{No-proj-emp})$$

$$(i) \text{Dept-Hu} \leftarrow \rho_{\text{no}}(\pi_{\text{onumber}}(\sigma_{\text{location} = \text{'Houston'}}(\text{Dept-locations})))$$

$$\text{Proj-Hu} \leftarrow \rho_{\text{no}}(\pi_{\text{onum}}(\sigma_{\text{location} = \text{'Houston'}}(\text{Project})))$$

$$\text{Proj-Hu-not-Dept} \leftarrow \text{Dept-Hu} - \text{Proj-Hu}$$

$$\text{Projects-Hu} \leftarrow \text{Project} \bowtie_{\text{onum} = \text{no}} \text{Proj-Hu-not-Dept}$$

$$\text{Emp-Project-Hu} \leftarrow \text{Works-ON} \bowtie_{\text{pno} = \text{Pnumber}} \text{Projects-Hu}$$

$$\text{Result} \leftarrow \pi_{\text{Fname, Lname, Address}}(\text{Employees} \bowtie_{\text{essn} = \text{ssn}} \text{Emp-Project-Hu})$$

$$(j) \text{Dept-managers(essn)} \leftarrow \pi_{\text{mgr-ssn}}(\text{Department})$$

$$\text{Dept-managers-NoDep} \leftarrow \text{Dept-managers} - \pi_{\text{essn}}(\text{Dependents})$$

$$\text{Result} \leftarrow \pi_{\text{Lname}}(\text{Employees} \bowtie_{\text{essn} = \text{ssn}} \text{Dept-managers-NoDep})$$

Q22.

TABLE T1

P	Q	R
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10 a 5

15 b 8

25 a 6

TABLE T2

A	B	C
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10 b 6

25 c 3

10 b 5

(a) $T_1 \bowtie_{T_1.P = T_2.A} T_2$

(b) $T_1 \bowtie_{T_1.Q = T_2.B} T_2$

P	Q	R	A	B	C
10	a	5	10	b	6
10	a	5	10	b	5
25	a	6	25	c	3

P	Q	R	A	B	C
15	b	8	10	b	6
15	b	8	10	b	5

(c) $T_1 \bowtie_{T_1.P = T_2.A} T_2$

(d) $T_1 \bowtie_{T_1.Q = T_2.B} T_2$

P	Q	R	A	B	C
10	a	5	10	b	6
10	a	5	10	b	6
15	b	8	NULL	NULL	NULL
25	a	6	25	c	3

P	Q	R	A	B	C
15	b	8	10	b	6
NULL	NULL	NULL	25	c	3
15	b	8	10	b	5

$T_1 \bowtie_{T_1.P = T_2.A} T_2$

(e) $T_1 \cup T_2$

P	Q	R	A	B	C
10	a	5	10	b	6
10	a	5	10	b	6
15	b	8	NULL	NULL	NULL
25	a	6	25	c	3

P	Q	R
10	a	5
15	b	8
20	b	8
10	b	6
25	c	3
25	a	6
10	b	5

(f) $T_1 \bowtie_{(T_1.P = T_2.A \text{ AND } T_1.R = T_2.C)} T_2$

P	Q	R	A	B	C
10	a	5	10	b	5